

Nifty 100 Analytics - Analyst Guide

Sprint 6 Documentation

Days 36-45

Version 1.0.0

1. Project Purpose and Scope

End-to-end FinTech analytics platform for the Nifty 100 analytical universe.

Covers financial statements, ratios, screening, peers, sectors, clustering, valuation, REST APIs and

2. Data Foundation

SQLite database: nifty100.db

Core tables include companies, profitandloss, balancesheet, cashflow, financial_ratios, market_cap,

Sprint 6 analytical universe contains 92 companies.

3. Financial KPI Analysis

Key KPIs include ROE, debt-to-equity, operating margin, revenue CAGR, PAT CAGR, FCF CAGR, interest c
Latest-year analysis uses 2024 data where available.

4. Company Screener

REST endpoint: `/api/v1/screener`

Supports ROE, debt-to-equity, FCF, sector, revenue CAGR, PAT CAGR and P/E filters.

Invalid numeric parameters return HTTP 400.

5. Peer and Sector Analytics

Sector APIs provide company counts and median KPI values.

Peer APIs provide percentile comparisons across the defined peer groups.

Company peer comparison exposes 10 analytical metrics.

6. Clustering and Advanced Analytics

KMeans clustering uses 5 clusters and random_state=42.

Features: ROE, debt-to-equity, revenue CAGR, FCF CAGR and operating margin.

StandardScaler is applied after sector-median imputation.

Outputs: output/cluster_labels.csv and reports/elbow_plot.png.

7. REST API Guide

FastAPI service runs on port 8000.

Swagger documentation: /docs

OpenAPI specification: docs/openapi.json

Postman collection: docs/postman_collection.json

The API contains 16 Sprint 6 endpoints.

8. QA and Testing

Automated test suite contains 102 passing tests.

Performance test: 10 concurrent screener requests completed in 0.302 seconds.

Five company profile requests completed in 0.099 seconds.

FastAPI and Streamlit services were verified.

9. Dashboard Operations

Streamlit dashboard runs on port 8501.

Run with: `streamlit run src\dashboard\app.py`

Dashboard covers company profiles, screening, peers, trends, sectors, capital allocation, annual rep

10. Known Data Exceptions

The SQLite companies table contains 100 master records, while the Sprint 6 analytical universe contains 100. JIOFIN has only two years of data, so its annual tearsheet is intentionally skipped.

FCF CAGR is unavailable for several companies and clustering uses sector-median or overall-median for those companies.

11. Sprint 6 Sign-Off Preparation

Verify all 20 acceptance criteria.

Archive final deliverables under output/final_deliverables.

Record QA and performance evidence.

Prepare the acceptance checklist for team-lead review.